Core Research Questions

Version: 2.0
Date: 2026-06-06
Status: Expanded — Integrates CoreQuestions_vNext.md, SRT v0.2, OpenProblems v1.0, HANDOVER priorities

> Questions are organized by level of abstraction.
> Higher levels represent meta-questions about lower levels.
> Each level is linked to its connected hypotheses (H1–H10) and open problems (OP-01–OP-09).
> **No question below is resolved. These are organizing frameworks for investigation, not answers.**

Program-Level Central Question

> **How do humans come to see something new?**

This question frames the entire research program at its highest level of abstraction. It subsumes the theory-level central question:

> **How do humans detect explanatory significance?**

Both questions remain open. The second is the current operational focus of Paper 4.

The Question Hierarchy

The questions form a layered structure from empirical observation (Level 0) through structure and discovery (Levels 1–2), significance and questions (Levels 3–4), explanation (Level 5), and concept evolution (Levels 6–8).

Each level asks a different kind of "why" — the levels do not merely add detail; they shift the type of question being asked.

Level 0 — Observation

What empirical phenomena require explanation?

These are the ground-level observations that started the research program. They are factual starting points, not questions about psychology or cognition.

- **Q0.1** Why do XOR/XNOR Boolean functions produce checkerboard patterns in Karnaugh maps?
- **Q0.2** Why do symmetric Boolean functions generate recurring visual forms (ring, corner, point) in K-map representations?

- **Q0.3** Why does variable rearrangement produce visually distinct K-maps for the same logical function?
- **Q0.4** What structural properties of a Boolean function survive all 24 variable arrangements?

Primary case studies: Papers 1, 2, 3
Connected hypotheses: H2 (representation reveals structure)
Note: Level 0 questions have candidate answers in the empirical papers. They serve as the foundation that motivates all higher-level questions.

Level 1 — Structure
How do patterns become structures in human perception?

Level 0 describes what happens in mathematics. Level 1 asks what happens in the observer.

- **Q1.1** Why do humans recognize these patterns as structures rather than noise or coincidence?
- **Q1.2** What distinguishes a pattern from a structure in human perception?
- **Q1.3** Why is the checkerboard recognized as a single unified whole rather than a cell-by-cell aggregate?
- **Q1.4** Does structure recognition require prior conceptual knowledge, or can it operate without it?
- **Q1.5** Is structure recognition a perceptual act, a cognitive act, or both?

Connected hypotheses: H1 (structure discoverer), H2 (representation transformation)
Connected open problems: OP-01 (attention selection), OP-04 (representation design)
Relation to Level 0: Level 1 does not ask why the pattern exists (Level 0), but why humans notice and name it as a structure.

Level 2 — Discovery
Why do humans discover particular structures at particular times?

Level 1 asks how patterns become structures. Level 2 asks why some structures are discovered and others are not.

- **Q2.1** Why do humans discover particular structures and not others of equivalent visual complexity?
- **Q2.2** What determines which structures attract sustained investigative attention?
- **Q2.3** Is structure discovery a deliberate act of search, or a spontaneous recognition?
- **Q2.4** What role does prior conceptual knowledge play in enabling structure discovery?
- **Q2.5** If AI and a human examine the same Karnaugh map, do they discover the same structures?

Connected hypotheses: H1 (structure discoverer), H3 (attention filter)
Connected open problems: OP-01 (attention selection), OP-06 (AI structure discovery)
Relation to Level 1: Level 2 is not asking how patterns become structures, but why particular structures are selected for sustained attention.

Level 3 — Significance
Why do certain structures feel important?

Level 2 asks which structures attract attention. Level 3 asks why some attended structures feel worth investigating.

- **Q3.1** Why do certain structures feel important while others of equal clarity do not?
- **Q3.2** What are the operative criteria of the significance filter (H4)?
- **Q3.3** Is invariance under transformation a universal significance criterion, or is it domain-specific?
- **Q3.4** Does the significance judgment operate consciously, pre-reflectively, or both?
- **Q3.5** Can the significance filter be made explicit and taught as a skill?
- **Q3.6** Does the significance filter differ across researchers, or are there shared criteria?

Connected hypotheses: H4 (significance filter), H6 (explanatory significance), H7 (mature structure discoverer)
Connected open problems: OP-02 (significance boundary), OP-05 (invariance significance)
Relation to Level 2: Attention (Level 2) is a prior selection; significance (Level 3) is a further evaluation. A structure can attract attention without being deemed research-worthy.

Level 4 — Question Generation
How do structures become research questions?

Not all significant structures generate research questions. Level 4 asks about the transition from significance judgment to question formation.

- **Q4.1** Why do certain structures develop into research questions while others do not?
- **Q4.2** What distinguishes a structure that generates curiosity from one that does not?
- **Q4.3** Is research question generation a skill that can be trained and improved?
- **Q4.4** Does the research question originate in the structure, in the observer, or in their interaction?
- **Q4.5** What is the role of conceptual vocabulary in making a question formulable?
- **Q4.6** Can a question only be formed after the right concept is available?

Connected hypotheses: H4, H6, H7
Connected open problems: OP-02 (significance boundary), OP-08 (research question

origin)
Relation to Level 3: A structure can pass the significance filter without generating a specific research question. Question generation requires not just judgment of significance, but the formation of a specific inquiry.

Level 5 — Explanation
How do humans orient toward explanation before finding it?

Level 4 describes question formation. Level 5 asks about the anticipatory structure of the inquiry — the sense that an explanation exists before it is found.

- **Q5.1** How do humans detect explanatory significance — the sense that a structure should have a structural explanation?
- **Q5.2** Do humans anticipate the existence of an explanation before they discover it?
- **Q5.3** What triggers the sense that "there must be a reason for this"?
- **Q5.4** Is explanation anticipation a separate cognitive act from structure recognition?
- **Q5.5** Can explanation anticipation occur without any prior theory about the domain?
- **Q5.6** Is explanation anticipation necessary for research initiation, or merely facilitative?

Connected hypotheses: H5 (explanation anticipation), H6 (explanatory significance)
Connected open problems: OP-03 (explanation anticipation mechanism)
Relation to Level 4: Question generation (Level 4) describes what the question is. Level 5 describes the motivational orientation — the felt sense that the structure demands an explanation — that drives the inquiry.

Level 6 — Concept
How does the formation of new concepts change what can be observed?

(Level 6–8 from CoreQuestions_vNext.md, integrated with SRT v0.2 H8–H10)

The levels above address what happens when a human encounters a phenomenon.
Levels 6–8 address how the existence of a concept changes what can be encountered in the first place.

- **Q6.1** What makes a concept effective as an observation lens (H8)?
- **Q6.2** Are concepts discovered or invented?
- **Q6.3** Why do some concepts generate new research programs while others remain merely descriptive?
- **Q6.4** Can a concept be deliberately designed to maximize its structure-revealing power?
- **Q6.5** Is the visibility-enhancing power of a concept (H8) measurable?

Connected hypotheses: H8 (concept as lens), H10 (generative concept)
Connected open problems: OP-07 (concept-as-lens)

Relation to Level 5: Levels 0–5 describe the processing of a single encounter with a phenomenon. Level 6 asks how having certain concepts changes what encounters are possible.

Level 7 — Expanded Perception
How does concept acquisition change what is visible?

Level 6 asks about the concept as a lens. Level 7 asks about the change in perception that results from acquiring the lens.

- **Q7.1** How do concepts change what a researcher can observe?
- **Q7.2** Why do some concepts reveal previously invisible structures?
- **Q7.3** Can new observations be produced by conceptual change alone, without new data or new instruments?
- **Q7.4** What is the relationship between conceptual change and observation change — do they always co-occur, or can one precede the other?
- **Q7.5** Is there a limit to how much a single concept can expand perception?

Connected hypotheses: H8 (concept as lens), H9 (concept evolution)
Connected open problems: OP-07 (concept-as-lens)
Evidence from this program: The concept of "structural invariance" made the question "what properties survive transformation?" formulable — a question that was not available before the concept existed.

Level 8 — New Discovery
How does the concept evolution cycle generate new discovery cycles?

Level 7 describes the change in perception due to a concept. Level 8 asks about the systemic dynamics — how a concept, by enabling new perception, generates new research, which generates new concepts.

- **Q8.1** What causes a research field or program to expand over time?
- **Q8.2** How do concepts generate further concepts through expanded perception?
- **Q8.3** Can the Concept Evolution Cycle (H9) be modeled formally?
- **Q8.4** Can the cycle be used deliberately — can a researcher accelerate their own discovery cycle?
- **Q8.5** Are there structural conditions under which the concept evolution cycle stalls or terminates?

Connected hypotheses: H9 (concept evolution), H10 (generative concept)
Evidence from this program: Pattern → Layer Structure → Equivalence → Structural Invariance → Structure Recognition Theory → Human-AI Collaboration Model. Each concept made the next visible.

Full Question Inventory

Questions are grouped by type and cross-referenced to levels above.

Foundational Questions (Level 1)

Q1. Why do humans discover structures?
(Connects to: H1, OP-01)

Q2. Why do humans perceive certain structures as non-random?
(Connects to: H5, OP-03)

Q3. Why do humans judge certain phenomena as meaningful?
(Connects to: H4, OP-02)

Q4. Where do research questions come from?
(Connects to: H6, OP-08)

Q5. How do humans and AI collaborate in the structure discovery process?
(Connects to: H1, H7, OP-06)

Extended Questions (Level 2)

Q6. Does something analogous to intuition exist in AI?
(Connects to: H1, H7, OP-06)

Q7. Does AI discover structures, or does it generate them?
(Connects to: H1, H7, OP-06)

Q8. Are structure discovery and meaning attribution the same process, or distinguishable?
(Connects to: H1, H4, H6)

Q9. What kind of cognitive profile characterizes a researcher?
(Connects to: H7 — researcher as explanation opportunity detector)

Q10. Will the future role of humans shift from computation toward structure discovery and significance evaluation?
(Connects to: H1, H7, OP-06)

Meta-Theoretical Questions (Level 3)

Q11. Can the significance filter (H4) be operationalized into a formal or computable model?
(Connects to: H4, OP-02)

Q12. Is explanatory significance detection (H6) a domain-general cognitive capacity, or domain-specific?
(Connects to: H6, H7)

Q13. What is the relationship between structural invariance (Papers 1–3) and explanatory significance?
(Connects to: H4, H6, OP-05)

Q14. Can concept generativity (H10) be measured or predicted before a concept is widely used?
(Connects to: H10, OP-07)

Q15. Does the Concept Evolution Cycle (H8–H9) have acceleration mechanisms — and if so, what are they?
(Connects to: H8, H9)

Cross-Level Questions

Some questions span multiple levels and cannot be cleanly assigned to one:

Question	Levels Spanned	Hypothesis Connections
What makes a structure feel "explanatorily pregnant"?	3 + 5	H4, H5, H6
Is there a unified cognitive mechanism underlying attention, significance, and explanation anticipation?	2 + 3 + 5	H3, H4, H5, H6
How does AI's role in discovery change as AI systems become more capable?	2 + 6	H1, H7, OP-06
Can the research generation cycle be deliberately initiated, or only recognized in retrospect?	1–5	H1–H7
Does the same question — "why does this pattern exist?" — operate differently in different domains?	0 + 3 + 5	H2, H4, H5

Program History as Question Evolution

The research program's central question has evolved through successive refinements. This evolution is itself an example of the Concept Evolution Cycle (H9):

Stage	Central Question	Conceptual Enabler
Initial	Why does this checkerboard appear?	Karnaugh map as visualization tool
Early	Why do humans recognize patterns as structures?	Distinction between pattern and structure
Middle	Why do humans discover structures?	Concept of discovery as distinct from recognition
Later	Why do certain structures feel worth investigating?	Concept of significance filter (H4)
Current	How do humans detect explanatory significance?	Concept of explanatory significance (H6)
Program-level	How do humans come to see something new?	Concept of concept as lens (H8)

What this table shows: Each new central question required a new concept to become formulable.
This is not incidental — it is the Concept Evolution Cycle demonstrating itself in the research program's own history.

Open Status Notice

All questions in this document are **open** unless explicitly marked otherwise.
No question should be marked as resolved without the researcher's explicit approval.

To add a new question:

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**Q[next number].** [Question text]

*(Connects to: [Hypotheses], [Open Problems])*
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To flag a question as having a candidate resolution, add:

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*Candidate resolution proposed — see [document reference]*
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Version history: v1.0 = initial structure (Level 0–5, Q1–Q10); v2.0 = expanded (Level 0–8, Q1–Q15, cross-level table, full hypothesis/OP connections, program history table)
Cross-reference: Hypotheses.md, OpenProblems.md, HANDOVER.md ("How Papers 1–3 Connect to SRT"), inha20/StructureRecognitionTheory_Unified.md (compressed SRT v0.3; full v0.3 text only in git 677bf7e^)
vNext candidate integrated: drafts/CoreQuestions_vNext.md (Levels 6–8 adopted)
Last Updated: 2026-06-06